WEEK-5

Monotonic → consistently increasing or decreasing
differentiable → diffentiable at point → if derivative exist
differentiable on an interval if it is differentiable at every point in that interval

If $y = g(x)$, then:

$$F_Y(y) = f_X\left(g^{-1}(y)\right) \cdot \left|\frac{d}{dy} g^{-1}(y)\right| \qquad g^{-1}(y) = \frac{1}{g(y)}$$

Q ① Let $X \sim Exp(\lambda)$. Find the PDF of X^2

↳ continous random variable

Sol.

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & , x > 0 \\ 0 & , \text{other} \end{cases}$$

$$Y = X^2$$

① $x = \sqrt{y}$ → inverse: $y = \sqrt{x}$ → $g^{-1}(x)$

② ~~$\frac{dx}{dy} = \frac{1}{2\sqrt{y}}$~~

$$g^{-1}(y) = \sqrt{y}$$

$$g^{-1}(y) = \frac{1}{2\sqrt{y}}$$

$$f_Y(y) = f_X\left(g^{-1}(y)\right) \cdot \left[g^{-1}(y)\right]'$$

$$\Rightarrow \lambda e^{-\lambda(\sqrt{y})} \cdot \frac{1}{2\sqrt{y}}$$

Q ② Let $X \sim \text{Uniform}([-3, 3])$. Find the PDF of $|X|$.

Sol.

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{other} \end{cases}$$

$$= \begin{cases} \frac{1}{6}, & -3 \le x \le 3 \\ 0, & \text{other} \end{cases}$$

$\boxed{\text{CDF}} \rightarrow F_Y(y) = P(Y \le y) = P(|x| \le y)$

$\hookrightarrow$ Integration: $F_Y(y) = \int_{-y}^{y} \frac{1}{6} \cdot dx = \frac{1}{6}[x]_{-y}^{y}$

$\Rightarrow \frac{1}{6}\left[y - (-y)\right] = \frac{y}{3}$

$\boxed{\text{PDF}} \rightarrow$ Use Differentiation

$\left(\frac{y}{3}\right)' \Rightarrow \boxed{\frac{1}{3}} \rightarrow$ Ans.

Q. ③ Let $X \sim \text{Exp}(\lambda)$. Find the PDF of $Y = (X)^3$

Sol. $Y = X^3$ $\quad \boxed{g(x)}$

$X = (Y)^{1/3}$ $\quad \boxed{g^{-1}(y)}$

- $g^{-1}(y)' = \frac{1}{3}(y)^{1/3 - 1} \Rightarrow \frac{1}{3(y)^{2/3}}$
- $f_Y(y) = f_X\left(g^{-1}(y)\right) \times \left[g^{-1}(y)\right]'$

$\boxed{f_X(x) = \lambda e^{-\lambda x}} \Rightarrow f_X\left(y^{1/3}\right) \times \frac{1}{3(y)^{2/3}}$

$\Rightarrow \cancel{f_X(y)} \; \lambda e^{-\lambda(y)^{1/3}} \times \frac{1}{3(y)^{2/3}}$

$\Rightarrow \boxed{\frac{\lambda}{3y^{2/3}}\left(e^{-\lambda y^{1/3}}\right)}$ $\quad : y > 0$

(4) Let x be a continuous random variable with PDF

$$f_x(x) = \begin{cases} x/12 & 1 < x < 5 \\ 0 & \text{otherwise} \end{cases}$$

find the PDF of $y = 2x - 3$.

Sol.

$$\boxed{y = 2x - 3} \rightarrow g(x)$$

$$g^{-1}(y)$$

$$\boxed{x = \frac{y+3}{2}}$$

$$(g^{-1}y)' = \frac{1}{2}$$

$$f_y(y) = f_x\left(g^{-1}(y)\right) \cdot \left[g^{-1}(y)\right]'$$

$$= f_x\left(\frac{y+3}{2}\right) \cdot \left[\frac{y+3}{2}\right]'$$

$$\Rightarrow \frac{y+3}{12} \cdot \frac{1}{2}\left[(1) + 0\right] \cancel{= \frac{y+3}{}}$$

$$f_y(y) = \begin{cases} \frac{y+3}{48} & -1 < y < 7 \\ 0 & \text{otherwise} \end{cases}$$

Expectation/ of Continuous Random Variable
μ, σ^2

$\boxed{\text{Expectation}}$: $E[g(x)] = \int g(x) \cdot f(x)\, dx$

expected value of $g(x)$

mean/μ

- $E[x] = \int_{-\infty}^{\infty} x\, f(x)\, dx$

- $var(x) = E(x^2) - (E(x))^2$

Distribution in CRV:

1. Uniform $(a, b) \rightarrow E(x) = \frac{a+b}{2}$, $Var(x) = \frac{(b-a)^2}{12}$
2. Exponential $(\lambda) \rightarrow E(x) = \frac{1}{\lambda}$, $Var(x) = \frac{1}{\lambda^2}$
3. Normal $(\mu, \sigma^2) \rightarrow E(x) = \mu$, $var(x) = \sigma^2$

Q. The PF of time x (in min) b/w calls at the customer care centre is given by:

$$f(x) = \begin{cases} \frac{1}{4} \cdot e^{-x/4} & \text{if } 0 \le x < \infty \\ 0 & \text{otherwise} \end{cases}$$

Find Expected time (in min) b/w calls.

Sol.

$$E(x) = \int_{-\infty}^{\infty} x \cdot f(x) \cdot dx = \int_{0}^{\infty} x \cdot \frac{1}{4} \cdot e^{-x/4} \, dx$$

$$\Rightarrow \frac{1}{4} \int_{0}^{\infty} x \cdot e^{-x/4} \, dx$$

$$\Rightarrow \frac{1}{4} \cdot 16 \Rightarrow 4$$

$$E(x) = \frac{1}{\lambda}$$

$$\lambda = \frac{1}{E(x)} = \frac{1}{4}$$

Q. Let x be a continuous random variable. Probability that X takes a value within 2 standard deviation of the expected value is:

Sol.

- $P(|X-\mu| \ge k\sigma) \le \frac{1}{k^2}$ ↳ Ans. at least $\frac{3}{4}$

  ↳ This inequality gives the prob. that x is far from the μ (outside $k\sigma$)

- Substitute $k = 2$

  $P(|X-\mu| \ge 2\sigma) \le \frac{1}{2^2} = \frac{1}{4}$ → Prob. that x is outside 2σ is at most $1/4$

- Convert to "within 2σ"

  $P(|X-\mu| \le 2\sigma) = 1 - P(|X-\mu| > 2\sigma)$

  So: $P(|X-\mu| \le 2\sigma) \ge 1 - \frac{1}{4} = \frac{3}{4}$

Q. The PDF of a random var. X is given as:

$$f_X(x) = \begin{cases} x & 0 < x < 1 \\ 2-x & 1 \le x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the expected value of X^2.

Sol.

$$E(x^2) = \int_{-\infty}^{\infty} x^2 \cdot f(x)$$

$$E(x^2) = \int_{0}^{1} x^2 \cdot x + \int_{1}^{2} x^2 \cdot (2-x) \Rightarrow \frac{1}{4} + \frac{11}{12} \Rightarrow \frac{7}{6} \approx \boxed{1.17}$$

Q. The PDF of a rand. var. X is given as:

$$f_X(x) = \begin{cases} 2(1-x) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the variance of X.

→ If its like $x > 100$ then its: $\int_{100}^{\infty}$

Sol. $var(X) = E(X^2) - (E(X))^2$

$$E(x) = \int_0^1 x \cdot f(x)\, dx = \int_0^1 x \cdot 2(1-x) = 2\int_0^1 (x - x^2) = \frac{1}{3}$$

$$E(x^2) = \int_0^1 x^2 \cdot 2(1-x)\, dx = \frac{1}{6}$$

$$var(x) = \frac{1}{6} - \left(\frac{1}{3}\right)^2 \Rightarrow \frac{1}{18} = 0.06$$

Joint Distribution of Continous Random variable

- Joint PDF → $f_{XY}(x, y)$
- ☆ Conditional ~~PDF~~ Density → $f_{X|Y}(x|y) = \dfrac{f_{XY}(x,y)}{f_Y(y)}$ → Joint Dens / Marginal dens
- ☆ $f_Y(y) = \int f_{XY}(x, y)\, dx$ ⇐ [Marginal Density]
- ☆ Joint Density ⇒ $f_{XY}(x, y) = \dfrac{1}{|D|}$ → Area under curve

~~If Conditional Probability of Discrete~~

Q. Let $X \sim$ Uniform. Let $(Y|X=1) \sim Exp(1)$, $(Y|X=2) \sim Exp(2)$ and $(Y|X=3) \sim Exp(3)$. Find the distribution of Y.

Sol. $X \sim$ Uniform : $P(X=1) = P(X=2) = P(X=3) = \frac{1}{3}$

$(Y|X=1) \sim \lambda = 1$

$(Y|X=2) \sim \lambda = 2$

$(Y|X=3) \sim \lambda = 3$

~~Mixture~~

$f(y) = \lambda e^{-\lambda y}$

- $f_{Y|X=1}(y) = e^{-y}$
- $f_{Y|X=2}(y) = 2e^{-2y}$
- $f_{Y|X=3}(y) = 3e^{-3y}$

- Apply Mixture formula:

$$f_Y(y) = \frac{1}{3}(e^{-y}) + \frac{1}{3}(2e^{-2y}) + \frac{1}{3}(3e^{-3y})$$

$$= \frac{1}{3}\left[e^{-y} + 2e^{-2y} + 3e^{-3y}\right], \quad y > 0$$

Q.2 Let $X \sim \text{Binomial}\left(2, \frac{1}{2}\right)$. Let $(Y \mid X=0) \sim \text{Normal}\left(\overset{\mu}{0}, \overset{\sigma^2}{1}\right)$, $(Y \mid X=1) \sim \text{Normal}(1, 4)$ and $(Y \mid X=2) \sim \text{Normal}(2, 9)$. Choose the correct options from the following:

Sol.

- $P(X=0) = {}^2C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{2-0} \Rightarrow \frac{1}{4}$

  $P(X=1) = {}^2C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{2-1} \Rightarrow \frac{1}{2}$

  $P(X=2) = {}^2C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^0 \Rightarrow \frac{1}{4}$

- Normal PDF formula: $f(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$

$(y=2)$ Case 1: $\boxed{X=0} \longrightarrow \mu=0,\ \sigma=1$

$$f(2) = \frac{1}{\sqrt{2\pi}} \cdot e^{-2}$$

Case 2: $\boxed{X=1} \rightarrow \mu=1,\ \sigma=2$

$$f(2) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Case 3: $\boxed{X=2} \rightarrow \mu=2,\ \sigma=3$

$$f(2) = \frac{1}{3\sqrt{2\pi}} e^{0} = \frac{1}{3\sqrt{2\pi}}$$

~~Step~~ 0 Apply mixture:

$$f_Y(2) = \frac{1}{4}\left(\frac{1}{\sqrt{2\pi}} \cdot e^{-2}\right) + \frac{1}{2}\left(\frac{1}{2\sqrt{2\pi}} e^{-1/8}\right) + \frac{1}{4}\left(\frac{1}{3\sqrt{2\pi}}\right)$$

$$= \frac{1}{4\sqrt{2\pi}}\left[e^{-2} + e^{-1/8} + \frac{1}{3}\right]$$

Q. Let $X \sim \text{Bernoulli}(0.5)$. Let $(Y \mid X=0) \sim \text{Exp}(2)$ and $(Y \mid X=1) \sim \text{Normal}(0,1)$. Choose correct options.

Sol. $f_Y(y) \Rightarrow P(X=0) \cdot f_{Y|X=0} + P(X=1) \cdot f_{Y|X=1}$

$$\Rightarrow \frac{1}{\cancel{2}} \cdot \cancel{2} \cdot e^{-2y} + \frac{1}{2}\left(\frac{1}{1 \cdot \sqrt{2\pi}} \cdot e^{-1/2\left[\frac{y-0}{1}\right]^2}\right)$$

$$\Rightarrow e^{-2y} + \frac{1}{2\sqrt{2\pi}} \cdot e^{-\frac{1}{2}(y^2)}$$

$$P(x=1 \mid y=1) = \frac{P(x=1,\ y=1)}{P(y=1)}$$

Bayes formula:

$$\Rightarrow \frac{P(x=1)\cdot f_{x|y=1}}{f_y(y)}$$

$$\Rightarrow \frac{\frac{1}{2}\cdot\frac{1}{\sqrt{2\pi}}\, e^{-\frac{(1)^2}{2}}}{e^{-2(1)} + \frac{1}{2\sqrt{2\pi}}\cdot e^{-\frac{(1)^2}{2}}} \Rightarrow \frac{\frac{e^{-1/2}}{2\sqrt{2\pi}}}{\frac{2\sqrt{2\pi}\, e^{-2} + e^{-1/2}}{2\sqrt{2\pi}}}$$

$$\Rightarrow \boxed{\frac{e^{-1/2}}{\sqrt{8\pi}\, e^{-2} + e^{-2}}}$$

$$P(x=1 \mid y=-1) \Rightarrow \frac{P(x=1,\ y=-1)}{P(y=-1)} \Rightarrow \frac{P(x=1,\ y=-1)}{\cancel{P(x=0,\ y=-1)}^{\,0} + P(x=1,\ y=-1)}$$

$$\Rightarrow 1$$

Q.(1.) HARD

Let $Y = XZ$, $X \sim \text{Uniform}\{-1, 0, 1\}$ and $Z \sim \text{Normal}(1, 2)$, where X and Z are independent.

So, Ans. $f_{X|Y=1}(-1) = \frac{e^{-1}}{1+e^{-1}}$

① $P(x=-1) = P(x=0) = P(x=1) = \frac{1}{3}$

② $\boxed{Y = XZ}$

Case: 1: $X=0 \rightarrow Y=0$

Case: 2 $X=1 \rightarrow Y = Z \sim N(1,2)$

Case: 3 $X=-1 \rightarrow Y = -Z \sim N(-1,2)$

③ Build Mixture:

$$f_Y(y) = \frac{1}{3} f_Z(y) + \frac{1}{3} f_{-Z}(y) + \frac{1}{3} f_{Y|X=0}(y)$$

(4) Normal PDF

$$f_z(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

$$\boxed{\begin{array}{l} \bullet\ \mu = 1 \\ \bullet\ \sigma^2 = 2 \end{array}}$$

$$f_z(y) = \frac{1}{\sqrt{4\pi}} e^{-\frac{(y-1)^2}{4}}$$

(5) Compute: $f_Y(1)$

$$f_z(1) = \frac{1}{\sqrt{4\pi}} e^0 = \frac{1}{\sqrt{4\pi}}$$

$$f_{-z}(1) = \frac{1}{\sqrt{4\pi}} e^{-\frac{(-1-1)^2}{4}} = \frac{1}{\sqrt{4\pi}} e^{-1}$$

$(y=-z)$

$$f_{z=0} = 0$$

$(x=0)$

$f_Y(1)$ Ans. $\boxed{\frac{1}{3\sqrt{4\pi}}(1+e^{-1})}$

For getting: $f_{X|Y=1}(-1) = \frac{e^{-1}}{1+e^{-1}}$

- Using Bayes:

$$P(X=x \mid Y=1) = \frac{P(X=x) \cdot f_{Y|X=x}(1)}{f_Y(1)}$$

Case $x=0 \rightarrow P(Y=1 \mid X=0) = 0$

So: $\boxed{f_{X|Y=1}(0) = 0}$ ✓

Case $x=1 \rightarrow P \propto \frac{1}{3} \cdot \frac{1}{\sqrt{4\pi}}$

Case $x=-1 \rightarrow P \propto \frac{1}{3} \cdot \frac{1}{\sqrt{4\pi}} e^{-1}$

Normalize: $f_Y(1) = \frac{1}{3\sqrt{4\pi}}(e^{-1}+1)$

$$\boxed{P(X=-1 \mid Y=1) = \frac{e^{-1}}{1+e^{-1}}}$$